* UNIT - 5 : Introduction to Power System *

* Power system :→

→ A power systems study is made up of various engineering analysis investigations.

→ The goal of each study is to have a safe, efficient and reliable power system for your facility under both normal and abnormal conditions.

* Importance of power system :→

→ A power system comprises of the various sub-systems that include generation, transmission and distribution.

① To model or to execute per phase analysis of power system components.

② To monitor the voltage at different buses, real and reactive power flow between buses.

③ To plan future expansion of the current system.

④ To analyze the system under different fault conditions and based on different scenarios.

* Main parts of a power System: ⇒

① Power plant : ⇒ Generates electricity.

② Generators : ⇒ Rotating machines that convert mechanical energy into electrical energy.

③ Transmission lines : ⇒ Transport electricity from power sources to load centers.

④ Transformers : ⇒ Step up or down voltage levels for transmission and distribution.

⑤ Substations : ⇒ Step up or down voltage to meet requirements.

⑥ Distribution lines : ⇒ Distribute power where it's needed.

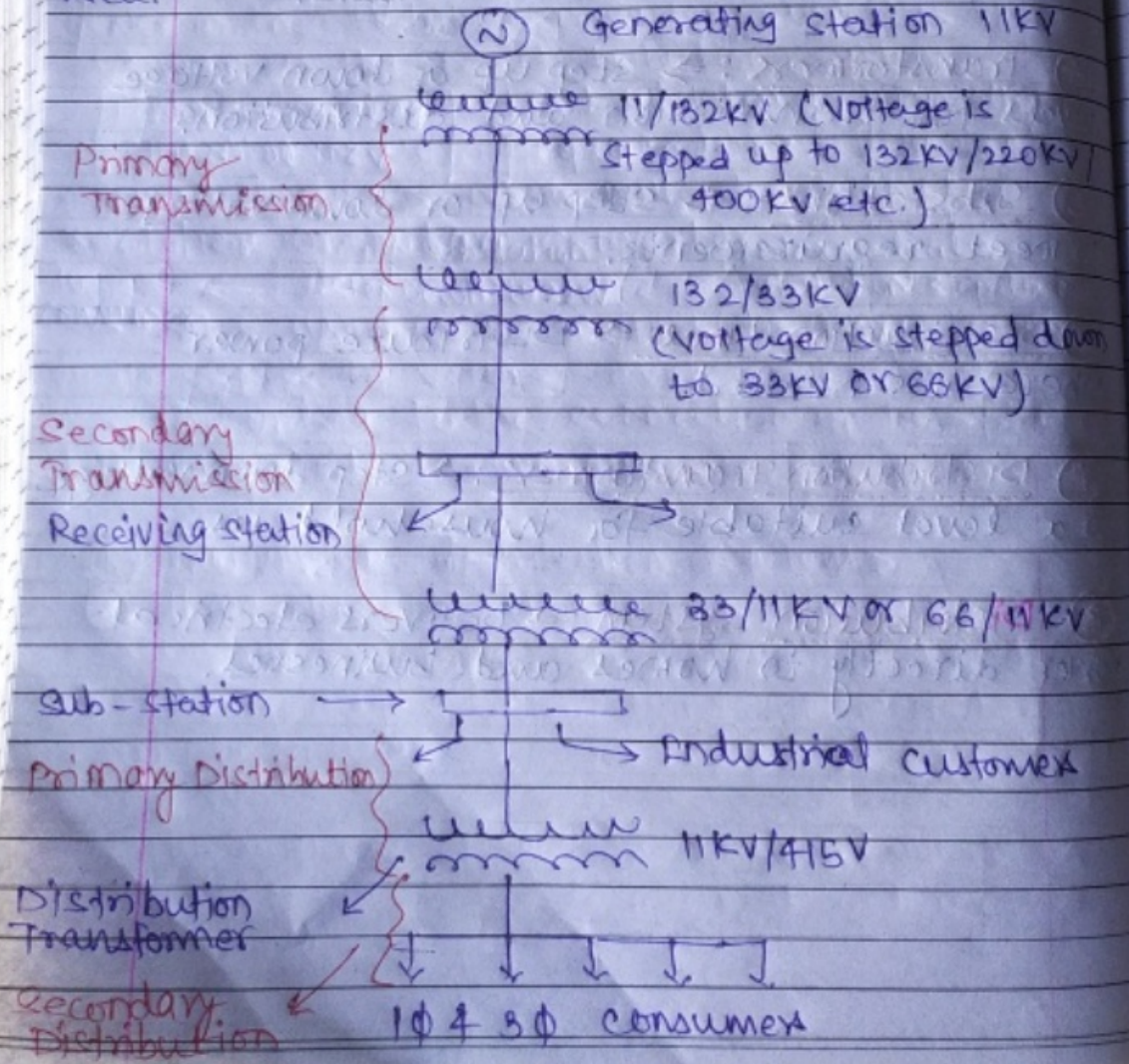
⑦ Distribution Transformers : ⇒ Step down voltage to a level suitable for household use.

⑧ Distribution system : ⇒ Delivers electrical power directly to homes and businesses.

* Single Line Diagram (SLD) :→

→ A single-line diagram is a simplified representation of an electrical system.

→ Symbols and lines are used to represent the nodes and connections in the system and electrical characteristics may be included as well.



* Types of transmission *

(A) Primary transmission :→

→ The electric power at 132 kV is transmitted by a 3-phase, 3-wire overhead system to the outskirts of the city.

→ Primary electrical transmission is the 1st stage in the power distribution process.

→ This stage involves transmitting electricity from generating stations to substations.

→ To reduce transmission losses, the voltage level is boosted significantly during this stage.

→ Typically, transmission voltage is considered high voltage, usually above 50,000 Volts.

(B) Secondary Transmission :→

→ Secondary electrical transmission, on the other hand, occurs after the primary transmission.

→ At this stage, the electricity has reached the distribution substations.

→ Here, the high voltage is stepped down to a much ~~lower~~ lower voltage.

→ The secondary distribution voltage is typically less than 2,000 volts.

→ For residential customers in the U.S. the voltage is usually stepped down to 120/240V.

~~Power Generating~~

Power Generating Station at 11KV → Grid substation Step up to 132KV → Main substation Step Down to 33KV → Intermediate substation step down to 11 KV → City substation Step Down to 220V → To consumer

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* Types of Distribution system *

Generation → Transmission → Distribution

→ There are two types of electric power:
① AC power & ② DC power.

→ According to the types of power used in the distribution system, it's classified into AC distribution system and DC distribution system.

Ⓐ AC Distribution system :-

→ In most of the conditions, the power consumer or load requires AC power.

→ Therefore, the electric power is generated, transmitted and distributed in the form of AC power.

→ An AC power can be easily step up and step down with the help of transformer.

→ According to the voltage level, the AC distribution system is further classified into two types;

① Primary distribution system
② Secondary distribution system.

(B) DC Distribution system :→

→ Most of load connected to the power system is AC load.

→ But there is a certain application where we required DC - power.

→ To fulfill these applications, we use DC power in the distribution system and this system is known as the DC - distribution system.

→ In this case, generated AC power is converted into DC power with the help of a rectifier or rotary converter.

→ It is used in DC motors, Battery charging.

→ According to the connection of DC System, it is classified into two types;

1. Two Wire DC distribution system.
2. Three Wire DC distribution system.

* Substations & its types *

→ A substation is an electrical system with high-voltage capacity and can be used to control the apparatus, generators, electrical circuits etc.

→ The substations are mainly used to convert AC to DC.

* Types of substations

(1) Step up type substation :→

→ This type of substation gets the power supply from a near producing facility.

→ It uses a large power transformer for enhancing the voltage level for transmitting to the remote locations.

→ In this substation, the power transmission can be done by using a transmission lines.

(2) Step down substation :→

→ This type of substation is placed at different points in an electrical network.

→ They can connect different parts of the network and that are a source of distribution lines.

→ This type of substation can change the transmission voltage to a sub-transmission voltage (69 kV).

(3) Customer Substations :→

→ This type of substation works as the major source of power supply for specific business client.

(4) System stations :→

→ This substation includes the huge amount of power transfer across the station and it's known as "system station".

→ This stations only offer no power transformer while others do voltage exchange as well.

→ They are important to long term consistency.

(5) Distribution Type Substation :→

→ Distribution type substations are placed where the main voltage distributions are stepped-down to supply voltage.

→ The voltage of any two phases will be 400 V and the voltage between neutral and any phase will be 230 V.

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⑥ 11KV substations :→

→ The main purpose of the 11KV substation is to gather the energy which is transmitted at high-voltage from the producing station.

⑦ 220KV substation :→

→ Here, the 220KVA substation is the power capability utilized by the step-down transformer in the substation.

→ It has highest apparent power, a stepdown transformer can provide.

→ The received voltage level of this substation will be 220KV.

⑧ 132KV substation :→

→ The 132 KV substation is the rating of the step-down transformer, which has 132KV primary voltage.

→ Here, the voltage has to be stepped-down.

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"Accept the challenges so that you may feel the Acceleration of victory".